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BLIND4001

Exercise 4

1. $f(x) = \frac{x}{x+1}$

$$f(x_0) = \frac{x_0}{x_0+1}$$

$$f'(x) = \frac{(x+1)(1) - (x)(1)}{(x+1)^2}$$

$$f'(x) = \frac{x+1-x}{(x+1)^2}$$

$$f'(x) = \frac{1}{(x+1)^2}$$

$$f''(x) = \frac{\cancel{(x+1)^2}(0) - (1)[2(x+1)]}{[(x+1)^2]^2}$$

$$f''(x) = \frac{-2(x+1)}{[(x+1)^2]^2}$$

$$f''(x) = \frac{-2\cancel{(x+1)}}{(\cancel{x+1})^4}$$

$$f''(x) = \frac{-2}{(x+1)^3}$$

$$2. f(x) = \frac{2x^2}{x^4}$$

$$f(x_0) = \frac{2x_0^2}{x_0^4}$$

$$f'(x) = \frac{(\cancel{x^4})(4x) - (2x^2)(-4x^3)}{(x^4)^2}$$

$$f'(x) = \frac{4x^5 - 8x^5}{x^8}$$

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$$f'(x) = \frac{-4x^5}{x^8}$$

$$f'(x) = -\frac{4}{x^3}$$

$$f''(x) = \frac{x^3 \cancel{(-4)}^0 - (-4)(3x^2)}{(x^3)^2}$$

$$f''(x) = -\frac{12x^2}{x^6}$$

$$f''(x) = \boxed{-\frac{12}{x^4}}$$

$$3. f(x) = \ln x^2$$

$$f(x_0) = \ln x_0^2$$

$$f'(x) = \frac{1}{x^2} \times 2x$$

$$f'(x) = \frac{2}{x}$$

$$f''(x) = \frac{(x)(0) - 2(1)}{(x)^2}$$

$$f''(x) = -\frac{2}{x^2}$$